

Nicholas Richardson

Curriculum Vitae

Institute of Applied Mathematics
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Research Interests

Audio & Signal Processing, Unsupervised Learning, and Tensor Factorizations

Education

- Sep. 2022 – **Doctor of Philosophy**, *University of British Columbia*, Vancouver, BC
(Apr. 2026) Institute of Applied Mathematics, supervised by Michael P. Friedlander
- Sep. 2020 – **Master of Mathematics**, *University of Waterloo*, Waterloo, ON
May 2022 Applied Mathematics Department, supervised by Giang Tran
- Sep. 2016 – **Bachelor of Mathematics**, *University of Waterloo*, Waterloo, ON
Apr. 2020 Mathematical Physics Major, Pure Mathematics Minor

Publications

- 2025 Joel Saylor, **Nicholas Richardson**, Naomi Graham, Robert Lee, and Michael P. Friedlander. Tracking cu-fertile sediment sources via multivariate petrochronological mixture modelling of detrital zircons. *Journal of Geophysical Research: Earth Surface* (Accepted September 2025).
- 2025 Naomi Graham, **Nicholas Richardson**, Michael P. Friedlander, and Joel Saylor. Tracing Sedimentary Origins in Multivariate Geochronology via Constrained Tensor Factorization. *Mathematical Geosciences*.
- 2023 Michael L. Waite and **Nicholas Richardson**. Potential Vorticity Generation in Breaking Gravity Waves. *Atmosphere*, 14(5):881.
- 2023 **Nicholas Richardson**, Hayden Schaeffer, and Giang Tran. SRMD: Sparse Random Mode Decomposition. *Communications on Applied Mathematics and Computation*.
- 2023 Lam Si Tung Ho, **Nicholas Richardson**, and Giang Tran. Adaptive group Lasso neural network models for functions of few variables and time-dependent data. *Sampling Theory, Signal Processing, and Data Analysis*, 21(2):28.

In Preparation

Nicholas Richardson, Noah Marusenko, and Michael P. Friedlander. Multiple Scale Methods for Optimization of Discretized Continuous Functions.

Teaching

- Sep. 2023 – **Instructor**, *University of British Columbia*, Vancouver, BC
(Apr. 2026)
- Intro to Computing: Python (Fall 2025, Winter 2026)
 - Calculus I (Fall 2023, Winter 2025)
- Sep. 2022 – **Teaching Assistant**, *University of British Columbia*, Vancouver, BC
Dec. 2024
- Calculus I (Fall 2024)
 - Calculus II (Winter 2023)
 - Calculus III (Fall 2022)
- Sep. 2022 – **Mathematics Learning Centre Tutor**, *University of British Columbia*,
Apr. 2024 Vancouver, BC
- June 2020 – **Mathematics & Physics Tutor**, *Self-Employed*
Dec. 2024 High School and Undergraduate Levels
- Jan. 2018 – **Teaching Assistant**, *University of Waterloo*, Waterloo, ON
Dec. 2021
- Mathematics of Music (Fall 2020, 2021)
 - Special Topics in Mathematical Connections: Math & Music (Fall 2021)
 - Complex Analysis (Winter 2020)
 - Calculus 1 & 2 (Fall 2019, Winter 2018, 2020, 2021)

Awards

Society of Industrial and Applied Mathematics

July 2024 Student Travel Award

Natural Sciences and Engineering Research Council of Canada

May 2023 Canadian Graduate Scholarship – Doctoral (330 awards nationwide)

May 2021 Canadian Graduate Scholarship – Master's (840 awards nationwide)

May 2019 Undergraduate Student Research Award (3 149 awards nationwide)

University of British Columbia

May 2022 Faculty of Science PhD Tuition Award

Sep. 2022 Four Year Doctoral Fellowship

Sep. 2022 President's Academic Excellence Initiative PhD Award

University of Waterloo

May 2021 President's Graduate Scholarship

Aug. 2020 K.D. Fryer Gold Medal

June 2020 Dean's Honours List With Distinction

May 2019 President's Research Award

Jan. 2019 Paul Berg Memorial Award

Jan. 2019 Arthur Beaumont Memorial Scholarship

- Sep. 2016 President's Scholarship of Distinction
 Sep. 2016 Alumni @ Microsoft Entrance Scholarship in Mathematics

Software

Computer Skills

Julia, Python, MATLAB, Quarto, Typst, $\text{\LaTeX}$, Adobe Software, Haskell

Published Code & Packages

- 2024 MatrixTensorFactor.jl, *Julia*
github.com/MPF-Optimization-Laboratory/MatrixTensorFactor.jl
 2023 SedimentSourceAnalysis.jl, *Julia*
github.com/njericha/SedimentSourceAnalysis.jl
 2022 Sparse Random Mode Decomposition, *Python*
github.com/GiangTTran/SparseRandomModeDecomposition
 2017 Drasil, *Haskell*
jacquescurette.github.io/Drasil

Conference Presentations & Posters

- July 2025 **Joint SIAM/CAIMS Annual Meetings**, *Montréal, QC*
 BlockTensorDecomposition.jl: An Optimization Framework for Constrained Tensor Factorization (Oral).
 July 2025 **CAIDA/TrustML Workshop — ICML Visitors @ UBC**, *Vancouver, BC*
 Unsupervised Signal Demixing with Tensor Factorization (Poster).
 July 2025 **Vision and Learning Workshop @ ICML**, *Vancouver, BC*
 Unsupervised Signal Demixing with Tensor Factorization (Poster).
 Dec. 2024 **CMS Winter Meeting**, *Richmond, BC*
 Density Separation with Tensor Factorization (Oral).
 Oct. 2024 **INFORMS Annual Meeting**, *Seattle, WA*
 Density Separation with Tensor Factorization (Oral).
 Sep. 2024 **West Coast Optimization Meeting**, *Vancouver, BC*
 Density Separation with Tensor Factorization (Oral).
 July 2024 **SIAM Annual Meeting**, *Spokane, WA*
 Density Separation with Tensor Factorization (Oral).
 Oct. 2023 **Biennial Meeting of SIAM Pacific Northwest Section**, *Bellingham, WA*
 Non-negative Tensor Decompositions for Unsupervised Learning (Oral).
 June 2021 **Canadian Applied and Industrial Mathematics Society**, *Waterloo, ON*
 Sparse Random Wavelet Representation & Decomposition (Poster).
 Runner up in annual poster competition

Other Presentations & Posters

*Presenting author

- Oct. 2025 Gabriel Bertolini*, **Nicholas Richardson**, Naomi Graham, Joel E. Saylor, Michael P. Friedlander. Sediment source distribution and mixing identification with multitype provenance dataset via tensor factorization. At *Geological Society of America Connects 2025* (Oral).
- Apr. 2025 Joel E. Saylor*, **Nicholas Richardson**, Naomi Graham, Robert G. Lee, Michael P. Friedlander. Endmember modelling of detrital zircon petrochronology data via multivariate Tucker-1 tensor decomposition. At *European Geosciences Union 2025* (Poster).
- Oct. 2024 Joel E. Saylor, **Nicholas Richardson**, Naomi Graham*, Michael P. Friedlander. Multivariate petrochronological characterization of sediment sources via constrained Tucker factorization. At *Geological Society of America Connects 2024* (Poster).
- Dec. 2023 Michael L. Waite* and **Nicholas Richardson**. Viscous and diffusive generation of potential vorticity in stratified turbulence. At *American Geophysical Union 2023* (Invited Speaker).

Theses

- (2026) Computational Techniques for Analysis of Mixed Signals (PhD Thesis).
- 2022 A Sparse Random Feature Model for Signal Decomposition. *UWSpace*. (Master's Thesis).

Other Research

- May 2019 – **Computational Fluids Research Assistant**, *University of Waterloo*,
Aug. 2019 Waterloo, ON, supervised by Michael L. Waite
- May 2017 – **Software Engineering Research Assistant**, *McMaster University*,
Aug. 2017 Hamilton, ON, supervised by Spencer Smith & Daniel Szymczak

Community & Leadership

- Sep. 2025 – **SIAM Student Chapter**, University of British Columbia
(Apr. 2026) ○ Co-organizer for the first UBC Applied Mathematics Meeting (Oct. 2025)
- Sep. 2024 – **Undergrad Research Supervisor**, University of British Columbia
Jan. 2025 Noah Marusenko, Multiple Scale Methods for Tensor Factorizations.
- Sep. 2023 – **Math Graduate Committee**, University of British Columbia
(Apr. 2026) American Mathematical Society Graduate Student Chapter
○ Co-organizer and Seminar Host (2024–)
○ Social Convenor (2023–)

- Sep. 2022 – **UBC A Cappella Club**, Vancouver, BC
- (Apr. 2026)
 - Live and Recording Sound Engineer (2024–)
 - Arranger (2023–)
- Sep. 2016 – **UW A Cappella Club**, Waterloo, ON
- Apr. 2022
 - Arranger (2017–2022)
 - Musical Director (2017–2020)
 - Chair (2018–2020)
 - Club President Award and Best Club Award*
(Waterloo Undergraduate Student Association, 2020)
- Oct. 2018 – **Canadian A Cappella Conference**, Waterloo, ON and Virtual
- Oct. 2021
 - Annual conference of workshops with over 100 singers from across the country
 - Lead Organizer (2019–2021)
 - Best Club Initiative* (Waterloo Undergraduate Student Association, 2019)
 - Travel Organizer (2018)